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1. Introduction

Episiotomy remains one of the most frequently performed obstetric procedures worldwide, despite a sustained, decades-long shift toward more selective and restrictive use [1]. The World Health Organization, in its technical guidance on care in normal birth, found no evidence supporting the liberal use of episiotomy and noted that a rate in the region of 10% for normal vaginal deliveries had been achieved without harm to mother or infant [2]; several recent institutional and population-based studies report rates substantially above this figure [3,4]. Reported prevalence still ranges widely across settings, from below 10% in countries following a sustained restrictive policy to rates exceeding 60% in several European countries, including Romania, reflecting persistent heterogeneity in obstetric practice [5]. Regardless of the exact local rate, episiotomy continues to represent one of the most common surgical interventions performed on women globally, and its wound-healing outcomes therefore carry substantial public health relevance. In a recent multicentre UK prospective cohort of over 2000 women with childbirth-related perineal trauma, wound infection was documented in 9.5% of those who had undergone episiotomy [6]. This figure should be read in its clinical context. Obstetric anal sphincter injury (OASI) is the perineal trauma category associated with the greatest risk of severe wound complications, and its management routinely incorporates prophylactic broad-spectrum antibiotics, stool softeners and structured follow-up, which attenuate the observed infection rate. Episiotomy, by contrast, is repaired without any comparable prophylactic package, and is one to two orders of magnitude more frequent. The clinical relevance of the episiotomy wound therefore derives from the combination of a non-trivial per-case complication rate and a very large denominator, not from any claim that it is the most severe form of perineal trauma.

While most episiotomy wounds heal without incident, a non-trivial proportion of women experience delayed healing, wound dehiscence, infection, hematoma, or persistent discharge, with reported infection rates of approximately 5–10% in large recent cohorts [6]. Beyond the immediate postpartum period, these complications are associated with chronic perineal pain, dyspareunia, and impaired sexual quality of life, with consequences that can extend for months after delivery and affect a woman's ability to care for her newborn, resume daily activities, and breastfeed comfortably [7,8]. Despite this burden, current management of the episiotomy wound remains largely supportive in nature: local hygiene measures, analgesia, and expectant observation, with no widely adopted intervention specifically designed to actively enhance the biological process of perineal wound repair [7,8].

Platelet-rich plasma (PRP) has emerged over the past two decades as a regenerative strategy across multiple surgical and musculoskeletal disciplines. Prepared by centrifugation of a patient's own peripheral blood, PRP yields an autologous concentrate enriched in platelets and the growth factors they release upon activation, including platelet-derived growth factor, transforming growth factor-beta, and vascular endothelial growth factor which are thought to accelerate angiogenesis, modulate inflammation, and promote organized tissue repair [9,10]. Since its early use in cardiac surgery in the late 1980s, PRP has been adopted in orthopedics, sports medicine, dermatology, and plastic and reconstructive surgery, with encouraging results in wound healing and scar reduction [11,12]. Because PRP is autologous, it carries a comparatively low risk of allergic reaction or transmissible infection relative to allogeneic or synthetic alternatives [12,13].